

Samuel Zhang

Applied AI Engineer & Product Builder

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PROFILE

Applied AI engineer and founder who turns ambiguous, domain-heavy problems into products people love. At Marsh, builds and evaluates ranking and document-intelligence systems for brokers; at COVERD, leads a company-aware voice-interview product. Previously delivered GROWMAT at Pfizer, a transformative platform saving 120+ analyst hours monthly through full-stack engineering, product ownership and stakeholder-led adoption.

CORE EXPERTISE

AI & ML: Multi-agent systems, RAG and retrieval, LLM and embedding evaluation, document intelligence, learning-to-rank, modelling and experiment design

Engineering: Python, Julia, SQL, TypeScript/React, PostgreSQL, Docker and CI/CD

Product: Customer discovery, rapid prototyping, stakeholder management, responsible AI and adoption

SELECTED EXPERIENCE

Marsh Risk | AI Intern – Applied AI & ML Thesis

May 2026 – Present

London, United Kingdom

- Structured a proof of concept for planned Broker Workbench integration around three evidence pillars: quantitative placement ranking, qualitative contract review and speculative, time-bounded market intelligence.
- Developed internal broker lead-matching decision engine for lead insurer selection, combining historical trading performance, wording quality and real-time appetite signals into transparent, ranked lead-market recommendations.
- Translated findings for Marsh Matching Companion Phase 3 stakeholders into transparent, human-in-the-loop lead recommendations supporting broker judgement, wider market consideration and client outcomes.

COVERD | Founder & Product Lead (part-time)

Mar 2026 – Present

London, United Kingdom

- Led customer discovery with four design partners and pivoted from candidate-side CV tooling to an AI interviewer that first learns how each company and role actually works.
- Selected a multi-agent, graph-based voice architecture seeded with priors from company interviews; agents update evidence-weighted beliefs during candidate interviews to choose role-specific follow-ups.
- Tested three architectures across 20 candidate interviews; candidates praised the selected system's intelligent follow-ups, question quality and ability to make them feel heard and valued.

Pfizer, Analytical R&D | Web Application Developer & Product Owner (part-time)

Oct 2024 – Apr 2026

London, United Kingdom

- Secured senior sponsorship, significant funding and department-wide adoption through live demonstrations; retained for a two-year part-time extension to own GROWMAT's roadmap.

Pfizer, Analytical R&D | Data Analyst Undergraduate

Sep 2023 – Aug 2024

Sandwich, United Kingdom

- Built and deployed GROWMAT over ten months, a seven-component capacity-planning platform used by 40+ employees across four teams, with wider impact across 80+ people.
- Cut capacity calculations from 12 minutes to under 60 seconds (12×), saving 120+ hours monthly and replacing weekly workbooks with live data and automated two-hourly refreshes.
- Engineered concurrency protection, daily backups and 180-day rollback; built live dashboards for project timelines, compliance, analytical performance and future workloads.
- Prototyped pharmaceutical solubility modelling with statistical thermodynamics and Julia; worked across Pfizer, MIT and Imperial to translate open-source research into a practical drug-development workflow.

EDUCATION

Imperial College London | *MSc AI Applications and Innovation – Predicted Distinction* **Sep 2025 – Sep 2026**

London, United Kingdom

Deep Learning, AI Safety, Innovation Management, Machine Learning in Medical Imaging, Machine Learning in Climate Change

King's College London | *BSc Chemistry with Biomedicine – First-Class Honours* **Sep 2021 – May 2025**

London, United Kingdom

Professional placement; Computational Chemistry, Molecular Biology, Chemical Biology, Organic Chemistry; Associate of King's College London

EARLIER EXPERIENCE, RESEARCH & PUBLIC SERVICE

King's College London | *Coding Series Tutor & Curriculum Designer* **Jan 2023 – Apr 2025**

London, United Kingdom

- Designed and delivered 20+ programming, data-analysis and introductory ML sessions for 80+ chemistry students, achieving 90% satisfaction and Royal Society of Chemistry endorsement.
- Organised a cross-industry data-science careers panel for 100+ students and mentored learners in using technical skills to widen their career options.

Royal Society / King's College London | *Summer Research Fellow* **Jun 2023 – Jul 2023**

London, United Kingdom

- Built GPU-accelerated GROMACS workflows for protein–membrane molecular simulations, improving computational performance by 70% and iterating experimental design from simulation results.

King's College London | *Undergraduate Research Fellow* **Jun 2022 – Jul 2022**

London, United Kingdom

- Engineered MATLAB, Python and Excel pipelines for rotational spectroscopy, reducing analysis time by 40% and creating an adopted workflow underpinning a publication in progress.

Singapore Civil Defence Force | *Commander's Personal Assistant / Sergeant* **Jul 2019 – Jul 2021**

Singapore

- Built MATLAB COVID-19 resource-planning models from public epidemiological data and automated emergency-activation attendance across legacy systems for 1,200+ personnel.
- Supported senior leaders in time-critical operations, balancing incomplete information, rapid prioritisation and accountability across large-scale personnel operations.

SELECTED TECHNICAL PROJECTS

- **Home Lab & Private AI Infrastructure** – Run 23 services across seven Proxmox/Docker servers, with 42 GB VRAM AI compute, cross-architecture CI, network security and 106 TB usable RAID storage.
- **Stock Market Simulation Engine** – Built Julia/SQL order matching, real-time WebSocket order-book updates and a responsive trading interface.

AWARDS & ADDITIONAL INFORMATION

Awards: King's Research Experience Award; Associate of King's College London (AKC); SCDF Service Excellence Award; SCDF 1st Division HQ Wall of Fame; EARCOS Global Citizenship Award

Languages: English and Mandarin (native/bilingual); Italian (elementary)

INTERESTS

Music	Competitive brass band euphonium player and member of a fourth-place UniBrass ensemble; performed in three musical-theatre productions, including one self-directed, written and composed.
Photography	Former professional wedding photographer, now photographing nature and friends for the joy of the craft, and the smile on people's faces.
Making	Builds computers and home infrastructure across 12 machines, with an increasingly noticeable electricity bill.
Beyond	Hiking, climbing, science fiction and travelling, making full use of a Singaporean passport.